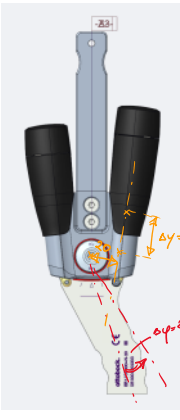
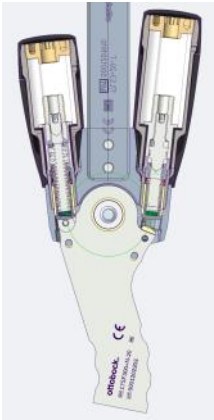


Tango Torque Calculation

Mittwoch, 25. Oktober 2023 19:25



17AD300=20

Assumption:

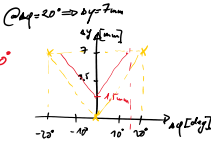
- For small ϕ $\rightarrow$ radiant = distance * angle

Radiant = $20\text{mm} \cdot 20\text{deg} = 7\text{mm}$ (approximately)

$\rightarrow$ therefore linear correlation (approximately)

Spring constant $R = 224\text{N/mm}$ (yellow spring, used by Patient Seeböth)

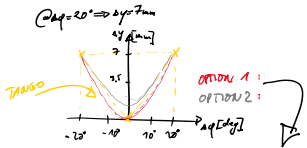
$\rightarrow$ Force_max = $224\text{N/mm} \cdot 7\text{mm} = 224 \cdot 7 = 1.568\text{N}$ (1,568kN) @ $\phi=20\text{deg}$



Code	D _H Hole Diameter	D _d Rod Diameter	L ₀ Free Length	R Spring Constant
	mm	mm	mm	N/mm
G 10 - 025			25	56.8
G 10 - 032			32	27.9
G 10 - 038			38	23.7
G 10 - 044			44	19.2
G 10 - 051			51	16.5
G 10 - 064			64	13.2
G 10 - 076			76	10.9
G 10 - 305			305	2.6
G 13 - 025			25	58.5
G 13 - 032			32	43.9
G 13 - 038			38	36.0
G 13 - 044			44	30.3
G 13 - 051			51	26.2
G 13 - 064			64	21.2
G 13 - 076			76	17.1
G 13 - 089			89	14.5
G 13 - 102			102	12.7
G 13 - 305			305	4.3
G 16 - 025			25	118
G 16 - 032			32	89.0
G 16 - 038			38	72.1
G 16 - 044			44	60.9
G 16 - 051			51	52.3
G 16 - 064			64	41.2
G 16 - 076			76	34.1
G 16 - 089			89	29.5
G 16 - 102			102	25.6
G 16 - 115			115	22.4
G 16 - 305			305	8.4
G 20 - 025			25	231
G 20 - 032			32	224

CAM-Design should:

- Reproduce max force @ $\phi=20\text{deg}$ in plantar/dorsalflexion
- Have a linear behavior like the tango in both directions for comparison purpose to the tango
- Be able to change pre-flexion for alignment purpose



- Option 1&2 have both smooth transition from dorsal to plantarflexion and viceversa
- After the zero position the steep of option 1&2 is close to the Tango-Spring ($R=224\text{N/mm}$)

